

Solution Requirements

Date	16 September 2026
Team ID	SWTID-2026-3342
Project Name	AI BlogNest API
Maximum Marks	4 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users register and log in with name, email & password; a JWT is issued on successful login and required on every protected route.	High
2	Authorization levels	Role-based access with four roles - <code>admin</code> , <code>editor</code> , <code>author</code> and the default <code>reader</code> - enforced via authentication & role middleware.	High
3	External interfaces	Integrates with Google Gemini AI for two dedicated flows: <code>/api/ai/generate-blog</code> (topic → full post) and <code>/api/ai/summarize</code> (content → concise summary).	High
4	Transactions / data processing	Full CRUD on Blog documents (title, content, category, tags, status, likes) and Comment documents, each linked to their owning User via a reference.	High
5	Reporting	Returns structured JSON for every blog list/detail call, including the populated author's name and email, suitable for any client to render.	Medium
6	Business rules	Each user's email must be unique; a new Blog defaults to <code>status: "draft"</code> until explicitly published; a new Comment defaults to <code>status: "pending"</code> until moderated by an Editor/Admin; only the owning Author (or an Admin) may edit or delete a given post.	High
7	Compliance to laws or regulations	Passwords are required with a minimum length and never returned in API responses; JWT secrets and the Gemini API key are kept in <code>.env</code> and excluded from version control via <code>.gitignore</code> .	High
8	Other	Centralized error-handling middleware (<code>errorMiddleware.js</code>) catches and formats failures from any route so the API never returns an unhandled stack trace.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Auth, Blog and Comment endpoints (MongoDB-only) should respond quickly; AI Generate/Summarize endpoints depend on Gemini's external latency.	Avg response < 300 ms (auth/blog/CRUD); AI endpoints measured separately (see Performance Testing)
2	Scalability	Stateless JWT authentication allows multiple Node.js instances to run behind a load balancer with no shared session state.	Designed to support growth beyond a single-instance deployment
3	Security & Data Privacy	Required/validated password field, JWT-based sessionless auth, and role middleware to block unauthorized access to admin-only routes.	No route accessible without a valid token where protection is required; all inputs validated
4	Reliability & Availability	Centralized error-handling middleware plus a <code>connectDB()</code> guard that exits the process cleanly if <code>MONGO_URI</code> is missing or the database is unreachable.	No unhandled crash on a bad request or a failed DB connection
5	Usability & Accessibility	Consistent, well-documented JSON REST responses, verified with a shared Postman / Thunder Client collection (<code>thunder-client-ai-blognest-api.postman_c...</code>).	Every endpoint documented and testable from the shared collection
6	Other (Maintainability)	Modular MVC-style structure (routes / controllers / middleware / models / services) with <code>morgan</code> request logging in development, so new endpoints can be added without touching existing modules.	New feature addable in a single new controller + route + (optional) service